

Question number	Answer	Mark
6(a)(i)	• Archaea	(1)

Question number	Answer	Mark
6(a)(ii)	A diagram that includes any three of the following structures: • circular DNA (1) • plasmid (1) • (70S) ribosomes (1) • membrane (1) • flagellum (1) • pili (1) • capsule (1)	(3)

Question number	Answer	Mark
6(b)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Indicative content • numbers of bacteria increase along the digestive system • <i>Veillonella</i> can tolerate only low pHs • <i>Streptococcus</i> found in pH 6 – 8 with reasonably high oxygen content • <i>Streptococcus</i> requires oxygen for aerobic respiration • <i>Enterobacterium</i> can tolerate low oxygen concentrations • <i>Enterobacterium</i> requires less ATP or can respire anaerobically • pH affects enzyme activity • pH affects the ionisation of R groups • small change in pH drastically affects enzyme activity	(6)

Level	Marks	Descriptor
	0	No awardable content.
1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.
2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning with some structure.
3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.